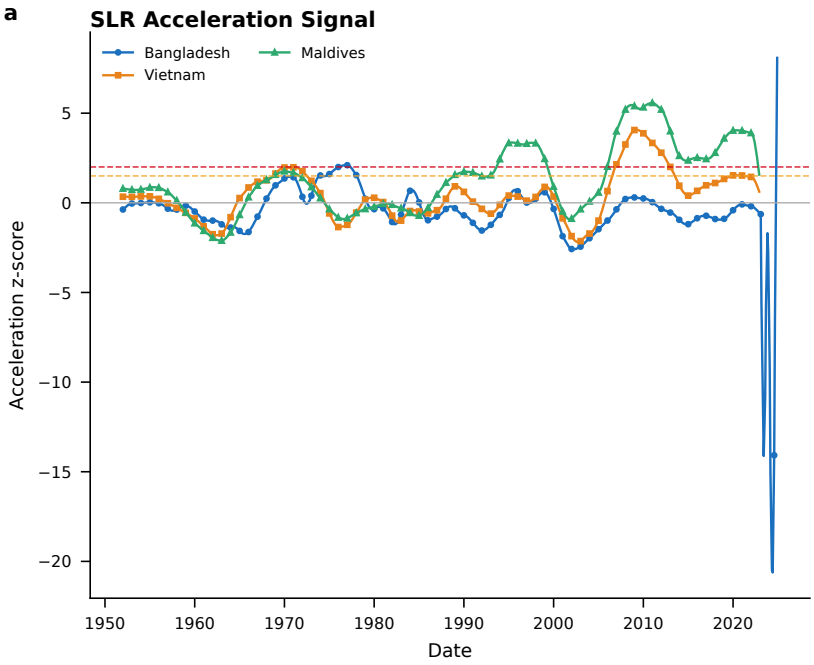
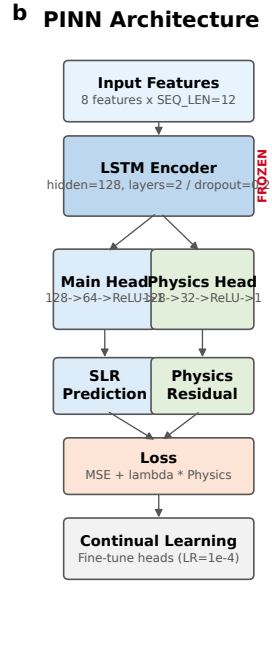

a
SLR Acceleration Signal
Bangladesh
Vietnam
Maldives
Acceleration z-score
5
0
−5
−10
−15
−20
1950
1960
1970
1980
1990
2000
2010
2020
Date


b
PINN Architecture
Input Features
8 features x SEQ_LEN=12
LSTM Encoder
hidden=128, layers=2 / dropout=
FROZEN
Main Head
128->64->ReLU->1
Physics Head
128->32->ReLU->1
SLR Prediction
Physics Residual
Loss
MSE + lambda * Physics
Continual Learning
Fine-tune heads (LR=1e-4)